

Operator Discovery Report: two_cluster_bottleneck_stagnation_perturbation_restart_controller

- Operator type: restart_controller.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministic stagnation-aware restart controller with optional perturbation restarts; triggers more often on difficult bottleneck-like instances while capping total restarts for runtime stability. Parameterized to trans
- Novelty classification: adaptive_restart_controller.
- Rediscovery signature: restart_controller:2:8:2.
- Surviving candidate: False.
- Problem class where it helps: two_cluster_bottleneck_tsp.

Pseudocode

- Start from a conservative restart budget.
- Increase restart count only after the configured stagnation trigger.
- Keep perturbation restarts bounded by a deterministic maximum.

Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: b9fd1976b1f4df02.

Why It Should Help

- The operator changes restart pressure under stagnation, which is a compact way to regulate exploration without rewriting search internals.

Failure Cases

- Does not transplant cleanly into sparse_three_opt.
- Does not transplant cleanly into clustered_local_search.
- Runtime inflation is high at 0.977422.

Transplant Results

- nearest_neighbor_2opt gap delta 0.0, runtime delta 146.953186, distance-eval delta 4830.285714.
- cheapest_insertion_2opt gap delta -0.002702, runtime delta 1276.115071, distance-eval delta 3566.285714.
- random_restart_2opt gap delta 0.0, runtime delta -243.858143, distance-eval delta -13398.285714.
- sparse_three_opt gap delta 0.035224, runtime delta 8.119657, distance-eval delta 32.142857.
- clustered_local_search gap delta 0.012072, runtime delta -105.229857, distance-eval delta -5890.0.

Instance-Family Results

- heldout_tsplib operator gap 0.235058 vs baseline 0.235058.

- clustered_tsp operator gap 0.110431 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.084632 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: -0.003304.
- Runtime inflation: 0.977422.
- Distance-evaluation inflation: 0.97364.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.003304.
- fixed_restart_budget mean gap 0.118013 and delta vs full operator 0.0.
- wider_restart_budget mean gap 0.117688 and delta vs full operator -0.000325.